# SIMATS ENGINEERING
# Saveetha Institute of Medical and Technical Sciences
# Chennai-602105

# CSA0927 - PROGRAMMING IN JAVA

## A Smart Event Management Framework for Real-Time Crowd Monitoring, Secure Entry, and Predictive Analytics

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# INTRODUCTION

- Large-scale events often face challenges such as overcrowding, long entry queues, security risks, and inefficient event planning.
- Traditional event management systems rely on manual processes, making them slow and error-prone.
- Smart technologies like QR-based ticketing, AI-driven crowd monitoring, and predictive analytics improve event operations.
- This framework integrates automation, real-time monitoring, and intelligent decision-making.
- It enhances visitor experience, organizer efficiency, and overall event safety.

# Problem Statement

- Manual ticket verification causes long waiting times.
- Lack of real-time crowd monitoring increases safety risks.
- Difficult to predict attendance and resource requirements.
- Inefficient communication between organizers and security teams.
- Limited insights into event performance after completion.
- Existing systems do not provide integrated planning, monitoring, and analytics.

# Objectives

- Develop a centralized smart event management platform.
- Enable secure QR-code based ticket verification.
- Monitor crowd density using AI and real-time analytics.
- Predict attendance and optimize resource allocation.
- Improve visitor safety through intelligent alerts.
- Provide interactive dashboards for event organizers.
- Enhance decision-making using predictive analytics.

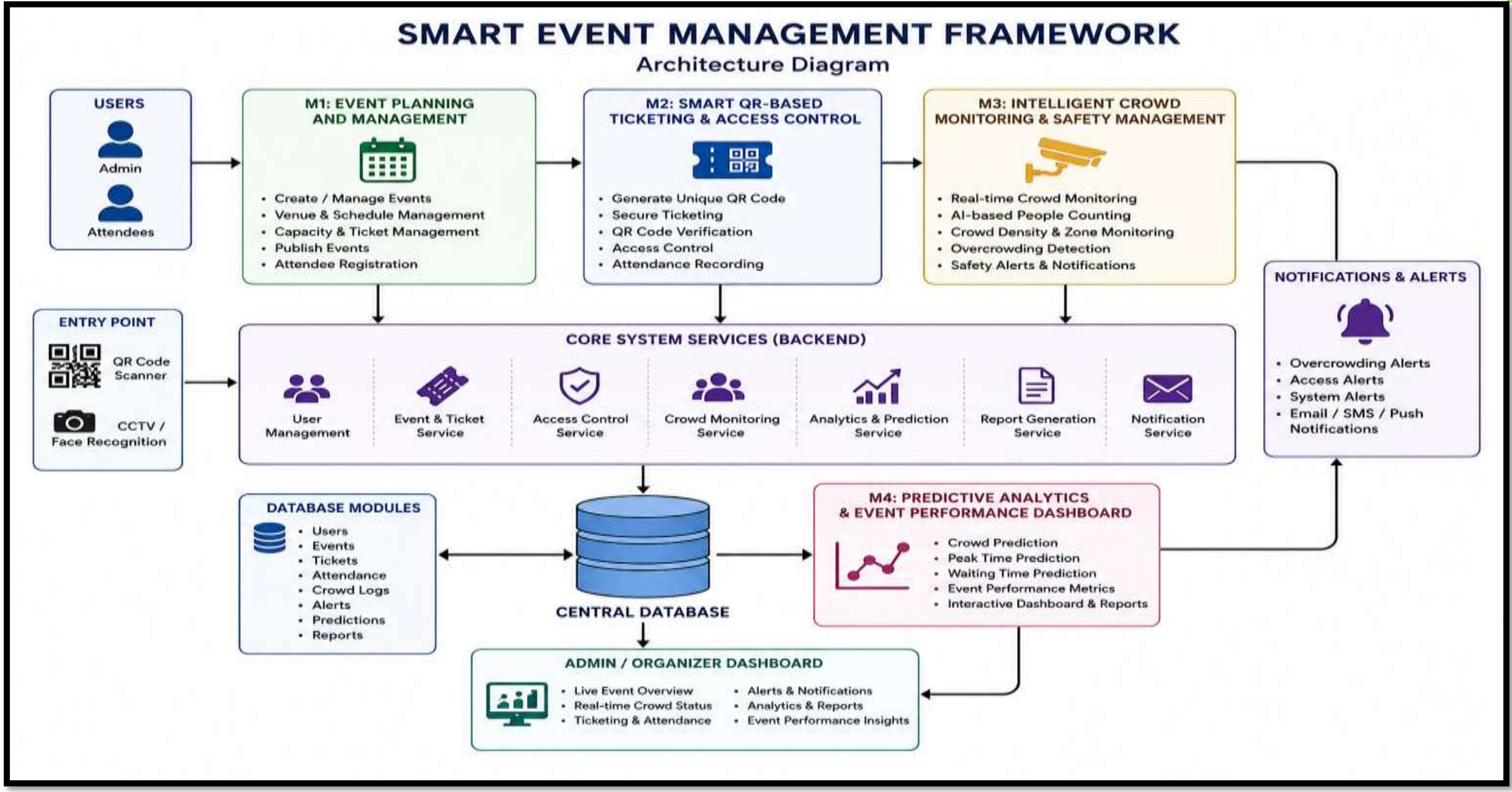
# Proposed System

- **Proposed System**
- The proposed system consists of four intelligent modules:
- Event Planning and Scheduling
- QR-Based Ticket Generation and Validation
- AI-Based Crowd Monitoring
- Predictive Analytics Dashboard
- **Features**
- Automated ticket management
- Secure QR authentication
- Real-time crowd monitoring
- Capacity alerts
- Attendance prediction
- Interactive dashboards

# Literature Review

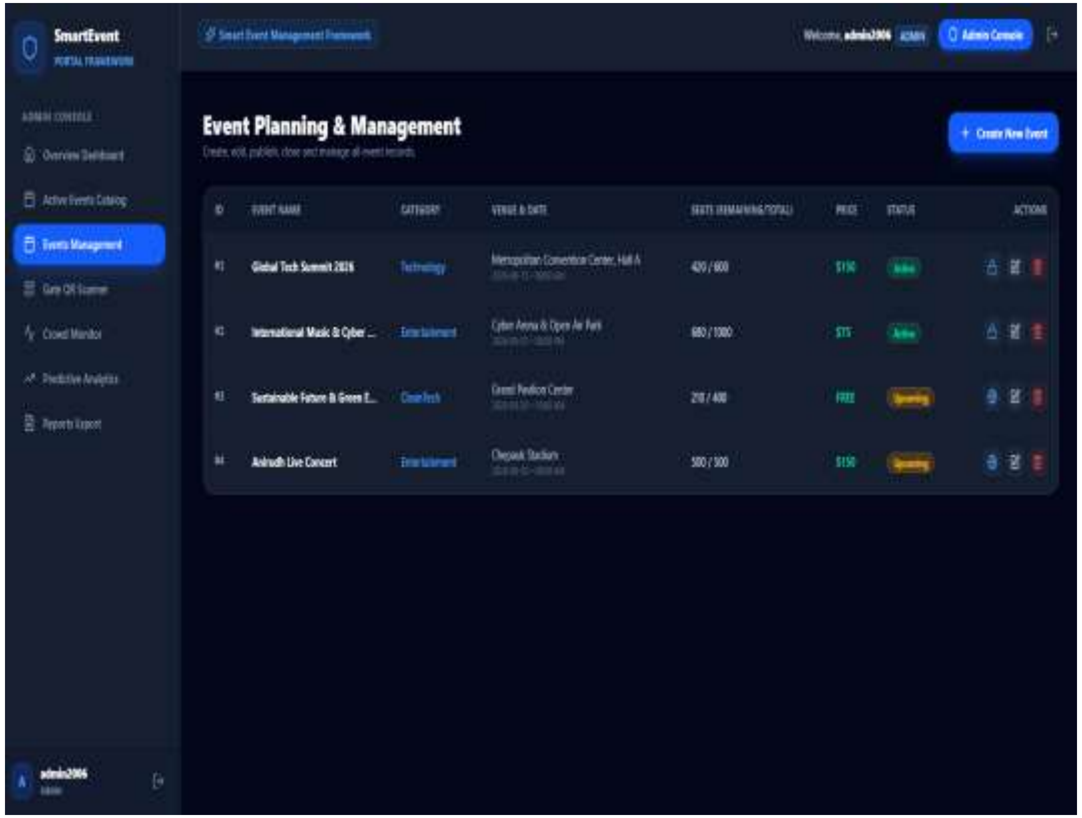
| S. No | Year | Author | Title | Journal | Methodology | Drawbacks |
|---|---|---|---|---|---|---|
| 1 | 2025 | Darsena et al. | Sensing Technologies for Crowd Management | IEEE Review | IoT, Crowd Sensing | Limited to transportation systems |
| 2 | 2024 | Zhou et al. | Understanding Crowd Behaviors in a Social Event | arXiv | WiFi Sensing, Data Mining | No ticketing or access control |
| 3 | 2024 | Ameen et al. | Advancements in Crowd-Monitoring Systems | arXiv | AI, Computer Vision | High computational cost |
| 4 | 2023 | collins | Event Management System with QR Code-Based Check-in | SJAIBT | QR Code, Cloud Computing | Limited predictive analytics |
| 5 | 2025 | Roy Choudhury et al. | Drishti AI-Event Guardian | arXiv | YOLOv8, Deep Learning | Focuses mainly on crowd monitoring |

# System Architecture



# Module 1 – Event Planning and Management

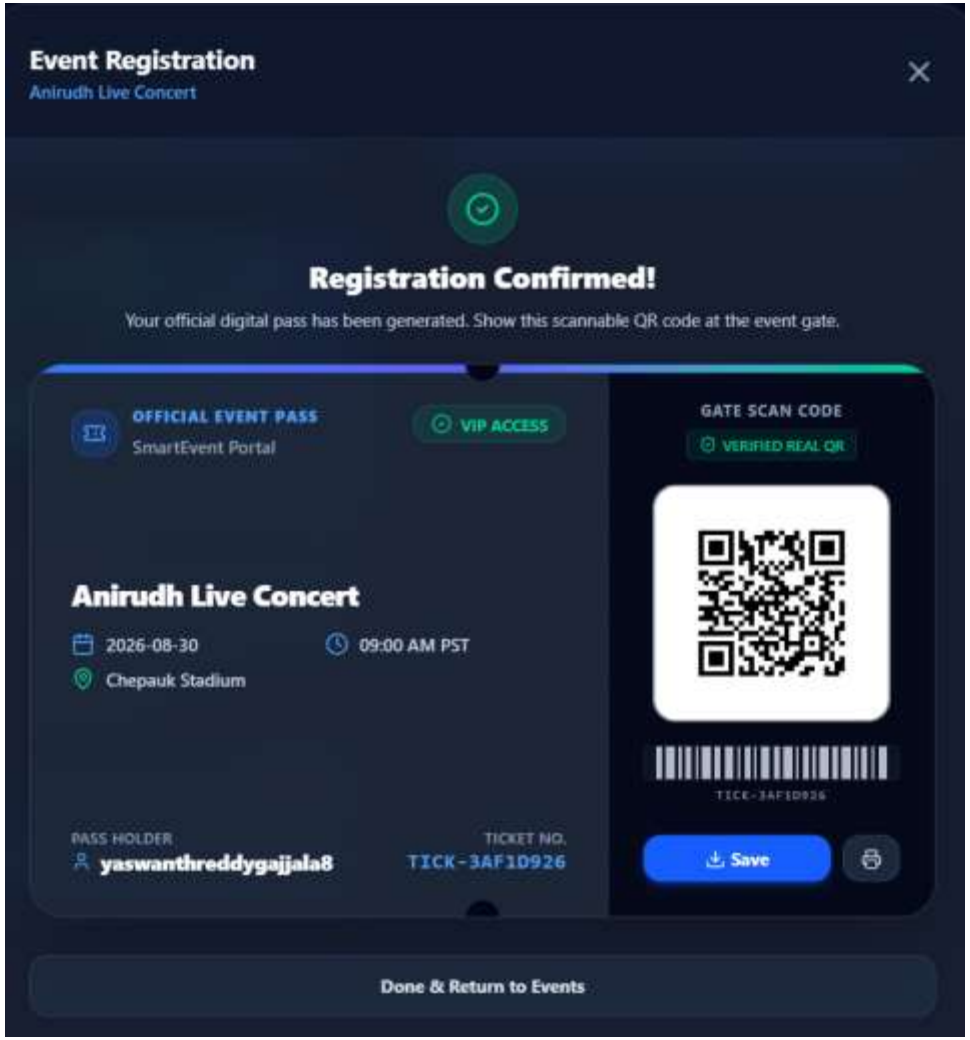
- Creates and schedules events with venue, date, and time details.
- Manages event resources such as staff, equipment, and vendors.
- Organizes event timelines and activity schedules.
- Allocates budgets and tracks event-related expenses.
- Sends notifications and updates to organizers and participants.
- Maintains centralized event information for smooth coordination.



# Module 2: Smart QR-Based Ticketing and Access Control

Engineer to Excel

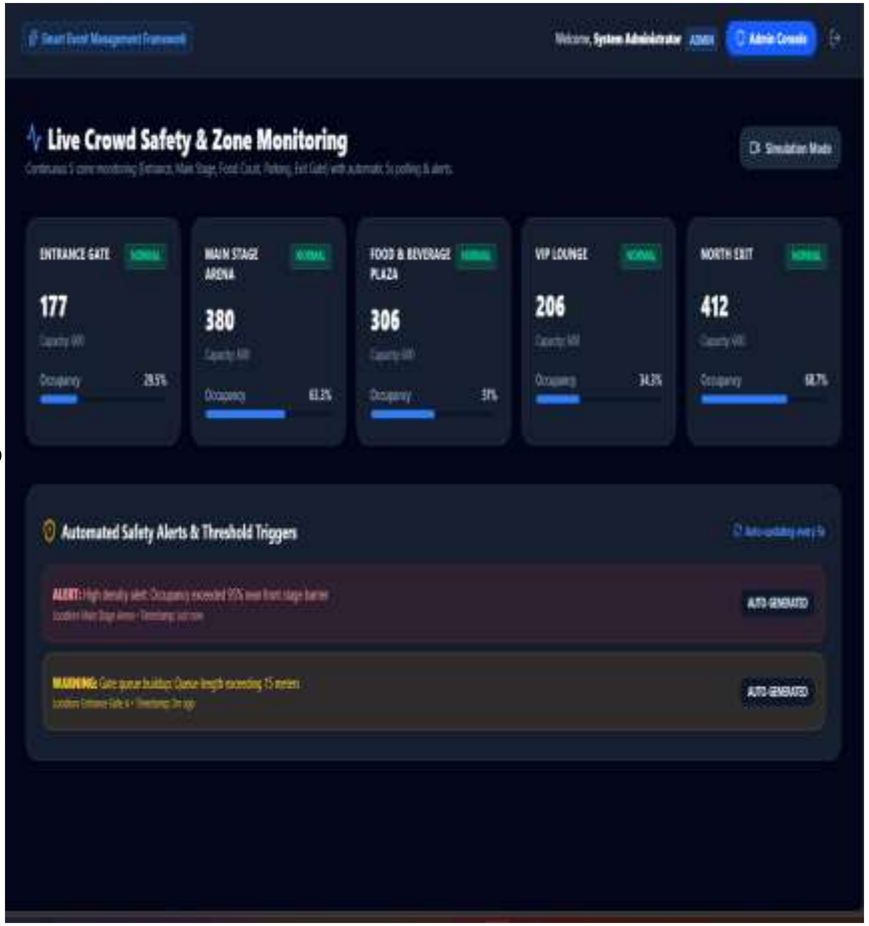
- Generates unique and secure QR-code tickets for attendees.
- Validates QR codes at entry points for fast authentication.
- Prevents duplicate entries and detects fraudulent tickets.
- Tracks real-time attendance and entry/exit records.
- Reduces waiting time through automated access control.
- Enhances security with digital verification and monitoring.



# Module 3: Intelligent Crowd Monitoring and Safety Management

Engineer to Excel

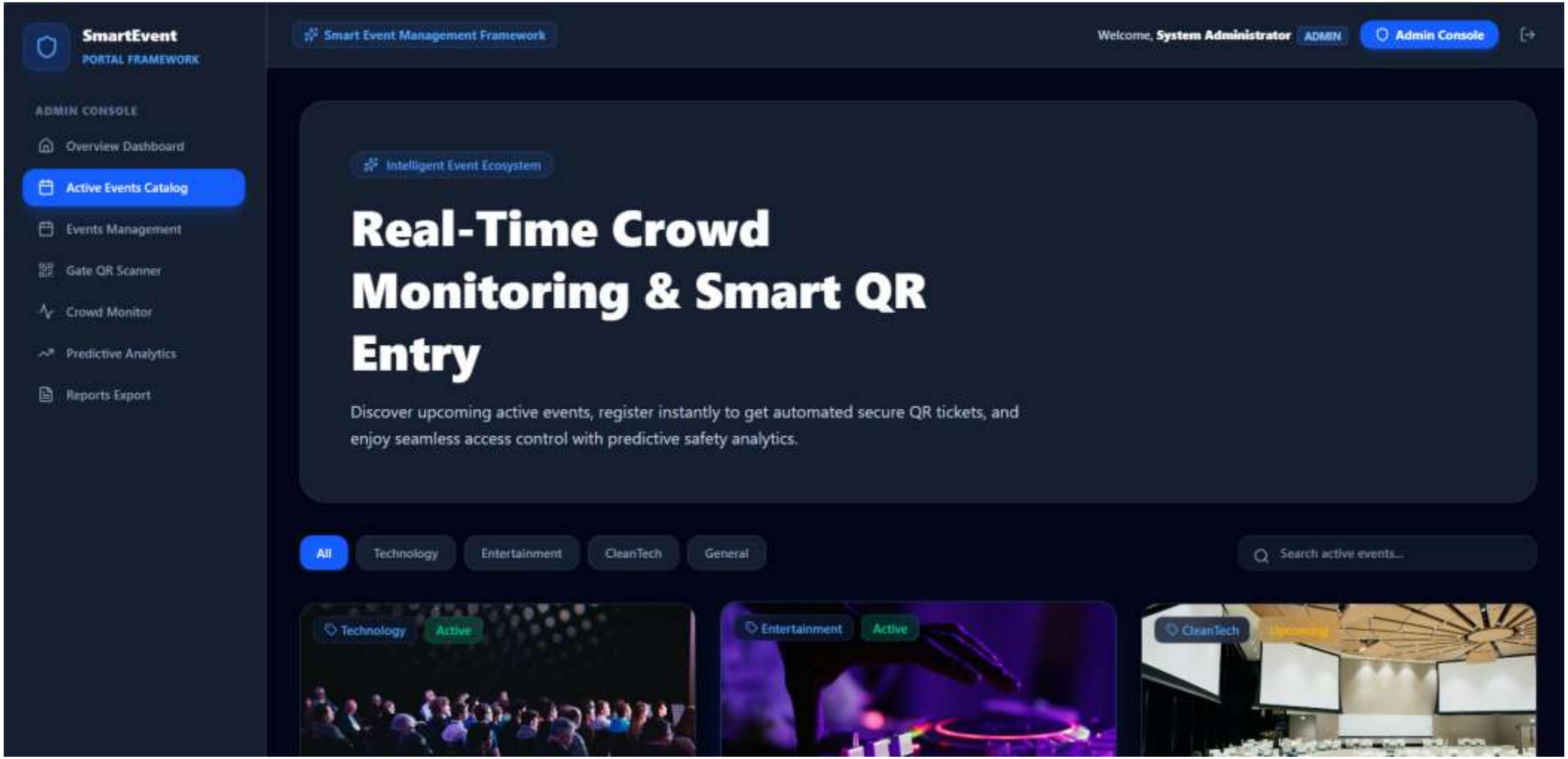
- Monitors crowd density using AI-enabled cameras and sensors.
- Detects overcrowding and unusual crowd movements in real time.
- Generates instant safety alerts during emergency situations.
- Assists security personnel with live crowd status updates.
- Visualizes crowd distribution through heatmaps and dashboards.
- Improves visitor safety by preventing congestion and accidents.



# Module 4: Predictive Analytics and Event Performance Dashboard

- Analyzes historical and real-time event data.
- Predicts attendance, resource requirements, and revenue trends.
- Provides interactive dashboards with key performance indicators (KPIs).
- Evaluates visitor engagement and event success metrics.
- Supports data-driven decision-making for future events.
- Generates detailed reports to improve event planning and management.

# Final output



# Challenges & Limitations

- **Challenges**
- Large-scale real-time data processing
- Camera placement limitations
- Network connectivity issues
- Data synchronization
- Privacy concerns
- **Limitations**
- Requires internet connectivity
- Camera quality affects crowd monitoring accuracy
- AI prediction depends on historical data quality
- High initial deployment cost

# Future Scope

- Facial recognition entry system
- IoT sensor integration
- Drone-based crowd surveillance
- Smart parking management
- AI chatbot support
- Multi-event management platform
- Digital payment integration
- Cloud-based scalable deployment

# Conclusion

- The proposed framework integrates planning, ticketing, crowd monitoring, and analytics into a single smart platform.
- AI and QR technologies improve operational efficiency and visitor safety.
- Predictive analytics supports better planning and decision-making.
- The system reduces manual effort and enhances the overall event experience.
- It provides a scalable solution suitable for concerts, exhibitions, conferences, and public gatherings.

# References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# THANK YOU